INORGANIC CUME

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A recent seminar on "Pt Nanoparticle Binding on Carbon Nanotube Fuel Cell Catalyst Surfaces" touched some basics and longstanding problems, listed in the following, in the fuel cell science:

(1). Alkaline fuel-cell, solid oxide fuel-cell, PEM fuel-cell, phosphoric acid fuel-cell, direct methanol fuel-cell, and molten carbonate fuel-cell are the six types that people have studied extensively. Which type(s) would use the Pt nanoparticle, and how would the nanoparticles behave in a fuel-cell of your choice? ***(20')***

(2). Single-walled carbon nanotubes (SWCNTs) from commercial sources are usually a mixture of metallic, semiconducting, and insulating forms. It is believed that a composite membrane, formed by blending the SWCNT's into the Nafion membrane (a DuPont polymer for conducting protons and blocking the polar solvent molecules' migration across the membrane in a fuel cell), may improve the PEM fuel cell performance. Which SWCNT form(s) should be used in the fuel cell, and why? Would the SWCNT-Nafion composite membrane help block polar solvent's cross-over, and why? ***(20')***

(3). Pt is a metal in cubic structure whose nanoparticles are usually surfaced by (001) and/or (111) crystal faces, while the SWCNT is surfaced by 6-member carbon-rings. Predict which Pt surface(s) may prefer to bind on the SWCNT, and explain why you believe so. (*Note: Your idea on this underexploited question may be appreciated by the seminar speaker and many others in the fuel-cell and SWCNT-mediated catalysis sciences*) ***(20')***

(4). It is believed that a trace amount of CO in the fuel gas stream may quickly deactivate the Pt-nanoparticle catalyst. Please use a chemical reaction to explain why so, and then suggest a chemical reaction or a procedure for reactivating the catalyst. (*Note: You suggestion may help solve this longstanding problem in the PEM fuel-cell development*) ***(20')***

(5). Predict how the sintering of catalytic nanoparticles would take place and affect the fuel cell's performance. Propose a method for minimizing the sintering of Pt-nanoparticles on the SWCNT. (*Note: Your proposal may be valuable to many studying the SWCNT-mediated catalyses*) ***(20')***